

ARID DESERT LANDFORMS / THAR DESERTIFICATION

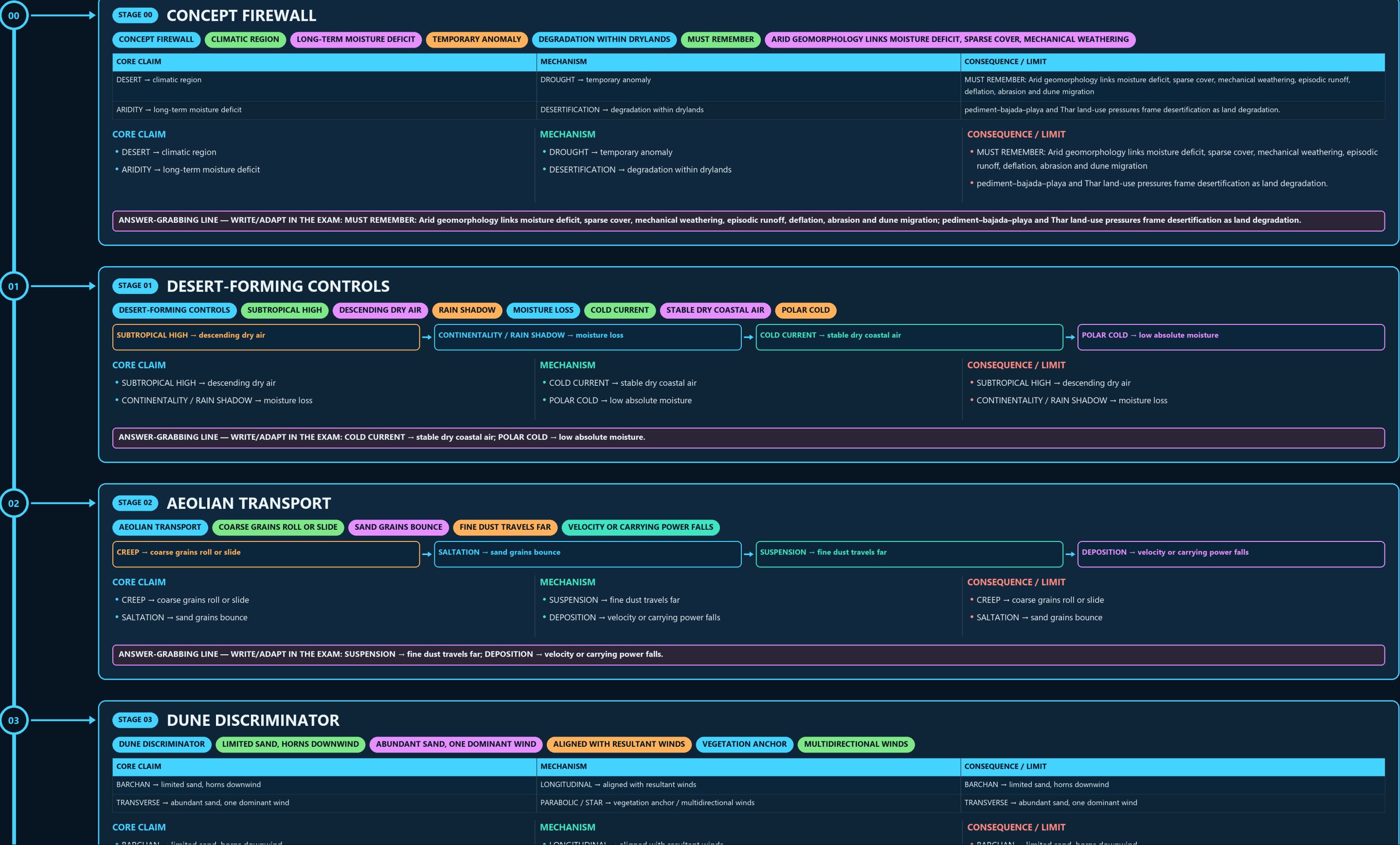
CONCEPT FIREWALL → DESERT-FORMING CONTROLS → AEOLIAN TRANSPORT → DUNE DISCRIMINATOR → WATER IN THE DESERT → THAR SPATIAL ANATOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

SALTATION → sand grains bounce

DEPOSITION → velocity or carrying power falls

SALTATION → sand grains bounce

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SUSPENSION → fine dust travels far; DEPOSITION → velocity or carrying power falls.

03

STAGE 03

DUNE DISCRIMINATOR

DUNE DISCRIMINATOR

LIMITED SAND, HORNS DOWNWIND

ABUNDANT SAND, ONE DOMINANT WIND

ALIGNED WITH RESULTANT WINDS

VEGETATION ANCHOR

MULTIDIRECTIONAL WINDS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
BARCHAN → limited sand, horns downwind	LONGITUDINAL → aligned with resultant winds	BARCHAN → limited sand, horns downwind
TRANSVERSE → abundant sand, one dominant wind	PARABOLIC / STAR → vegetation anchor / multidirectional winds	TRANSVERSE → abundant sand, one dominant wind

CORE CLAIM

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

MECHANISM

• LONGITUDINAL → aligned with resultant winds

• PARABOLIC / STAR → vegetation anchor / multidirectional winds

CONSEQUENCE / LIMIT

• BARCHAN → limited sand, horns downwind

• TRANSVERSE → abundant sand, one dominant wind

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PARABOLIC / STAR → vegetation anchor / multidirectional winds.

04

STAGE 04

WATER IN THE DESERT

WATER IN THE DESERT

STORM RUNOFF

WADI DISCHARGE

MOUNTAIN FRONT

ALLUVIAL FAN

FANS COALESCE

CLOSED BASIN

PLAYA AND SALT CONCENTRATION

STORM RUNOFF → wadi discharge

MOUNTAIN FRONT → alluvial fan

FANS COALESCE → bajada

CLOSED BASIN → playa and salt concentration

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• STORM RUNOFF → wadi discharge	• FANS COALESCE → bajada	• STORM RUNOFF → wadi discharge
• MOUNTAIN FRONT → alluvial fan	• CLOSED BASIN → playa and salt concentration	• MOUNTAIN FRONT → alluvial fan

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FANS COALESCE → bajada; CLOSED BASIN → playa and salt concentration.

05

STAGE 05

THAR SPATIAL ANATOMY

THAR SPATIAL ANATOMY

SANDY MARUSTHALI CORE

BAGAR AND ARAVALLI TRANSITION

LUNI AND SALINE-BASIN ROUTE

CANAL-ALLUVIAL TRANSFORMATION

EAST → Bagar and Aravalli transition

WEST → sandy Marusthali core

SOUTHWEST → Luni and saline-basin route

NORTH → canal-alluvial transformation

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• WEST → sandy Marusthali core	• SOUTHWEST → Luni and saline-basin route	• WEST → sandy Marusthali core
• EAST → Bagar and Aravalli transition	• NORTH → canal-alluvial transformation	• EAST → Bagar and Aravalli transition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOUTHWEST → Luni and saline-basin route; NORTH → canal-alluvial transformation.

06

STAGE 06

SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND

NATURAL STRESS

COVER LOSS + SOIL DISTURBANCE

WATER MISMANAGEMENT

SALINITY OR DEPLETION

THRESHOLD CROSSED

RECOVERY CAPACITY DECLINES

VARIABLE RAIN + WIND → natural stress

COVER LOSS + SOIL DISTURBANCE → exposure

WATER MISMANAGEMENT → salinity or depletion

THRESHOLD CROSSED → recovery capacity declines

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• VARIABLE RAIN + WIND → natural stress	• WATER MISMANAGEMENT → salinity or depletion	• VARIABLE RAIN + WIND → natural stress
• COVER LOSS + SOIL DISTURBANCE → exposure	• THRESHOLD CROSSED → recovery capacity declines	• COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

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06

STAGE 06 SUSCEPTIBILITY-TO-DEGRADATION

VARIABLE RAIN + WIND NATURAL STRESS COVER LOSS + SOIL DISTURBANCE WATER MISMANAGEMENT SALINITY OR DEPLETION THRESHOLD CROSSED RECOVERY CAPACITY DECLINES



CORE CLAIM

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

MECHANISM

- WATER MISMANAGEMENT → salinity or depletion
- THRESHOLD CROSSED → recovery capacity declines

CONSEQUENCE / LIMIT

- VARIABLE RAIN + WIND → natural stress
- COVER LOSS + SOIL DISTURBANCE → exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER MISMANAGEMENT → salinity or depletion; THRESHOLD CROSSED → recovery capacity declines.

07

STAGE 07 ARAVALLI BOUNDARY

ARAVALLI BOUNDARY MODIFIES LOCAL WIND AND RUNOFF DIVIDES AND ROUTES WATER SUPPORTS CONNECTIVITY NOT AN IMPERMEABLE DESERT WALL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
RELIEF → modifies local wind and runoff	HABITAT → supports connectivity	RELIEF → modifies local wind and runoff
DRAINAGE → divides and routes water	LIMIT → not an impermeable desert wall	DRAINAGE → divides and routes water

CORE CLAIM

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

MECHANISM

- HABITAT → supports connectivity
- LIMIT → not an impermeable desert wall

CONSEQUENCE / LIMIT

- RELIEF → modifies local wind and runoff
- DRAINAGE → divides and routes water

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HABITAT → supports connectivity; LIMIT → not an impermeable desert wall.

08

STAGE 08 CANAL BALANCE

CANAL BALANCE WATER SUPPLY AND PRODUCTION WATER-TABLE RISE POOR DRAINAGE WATERLOGGING AND SALINITY CROP-WATER FIT + DRAINAGE + MONITORING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
BENEFIT → water supply and production	POOR DRAINAGE → waterlogging and salinity	BENEFIT → water supply and production
SEEPAGE → water-table rise	MANAGEMENT → crop-water fit + drainage + monitoring	SEEPAGE → water-table rise

CORE CLAIM

- BENEFIT → water supply and production
- SEEPAGE → water-table rise

MECHANISM

- POOR DRAINAGE → waterlogging and salinity
- MANAGEMENT → crop-water fit + drainage + monitoring

CONSEQUENCE / LIMIT

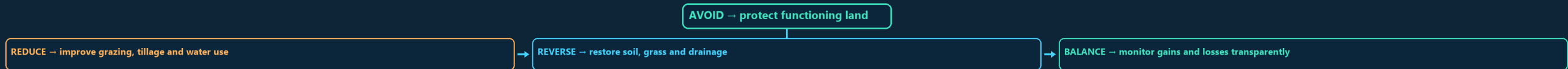
- BENEFIT → water supply and production
- SEEPAGE → water-table rise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGEMENT → crop-water fit + drainage + monitoring.

09

STAGE 09 LDN RESPONSE HIERARCHY

LDN RESPONSE HIERARCHY PROTECT FUNCTIONING LAND IMPROVE GRAZING, TILLAGE AND WATER USE RESTORE SOIL, GRASS AND DRAINAGE MONITOR GAINS AND LOSSES TRANSPARENTLY CLOSE DISTINCTION DESERT IS A CLIMATIC MOISTURE CONDITION, DESERTIFICATION IS DEGRADATION



CORE CLAIM

- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

MECHANISM

- REVERSE → restore soil, grass and drainage
- BALANCE → monitor gains and losses transparently

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly
- barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.

AVOID → protect functioning land

REDUCE → improve grazing, tillage and water use

REVERSE → restore soil, grass and drainage

BALANCE → monitor gains and losses transparently

CORE CLAIM

- AVOID → protect functioning land
- REDUCE → improve grazing, tillage and water use

MECHANISM

- REVERSE → restore soil, grass and drainage
- BALANCE → monitor gains and losses transparently

CONSEQUENCE / LIMIT

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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.

10

STAGE 10 RESTORATION MATRIX

RESTORATION MATRIX

WIND EROSION

COVER, SHELTER AND DUNE MANAGEMENT

RUNOFF EROSION

CONTOUR AND WATERSHED TREATMENT

DRAINAGE AND WATER BALANCE

RANGELAND DECLINE

NATIVE GRASS + INSTITUTIONS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
WIND EROSION → cover, shelter and dune management	RANGELAND DECLINE → native grass + institutions	different effects, so causal conclusions must be spatially qualified.
RUNOFF EROSION → contour and watershed treatment	EVIDENCE LIMIT: Wind is not the sole desert agent and every dune is not freely migrating	
SALINITY → drainage and water balance	grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have locally	

CORE CLAIM

- WIND EROSION → cover, shelter and dune management
- RUNOFF EROSION → contour and watershed treatment
- SALINITY → drainage and water balance

MECHANISM

- RANGELAND DECLINE → native grass + institutions
- EVIDENCE LIMIT: Wind is not the sole desert agent and every dune is not freely migrating
- grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have locally

CONSEQUENCE / LIMIT

- different effects, so causal conclusions must be spatially qualified.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have locally different effects, so causal conclusions must be spatially qualified.

11

STAGE 11 DESERTIFICATION ANSWER SPINE

DESERTIFICATION ANSWER SPINE

RETAIN UNCCD DRYLAND BOUNDARY

LOCATE PROCESS AND PRESSURE

SUSCEPTIBILITY PLUS ACCELERATOR

PROCESS-MATCHED, MONITORED AND QUALIFIED

DEFINE → retain UNCCD dryland boundary

MAP → locate process and pressure

DIAGNOSE → susceptibility plus accelerator

RESPOND → process-matched, monitored and qualified

CORE CLAIM

- DEFINE → retain UNCCD dryland boundary
- MAP → locate process and pressure

MECHANISM

- DIAGNOSE → susceptibility plus accelerator
- RESPOND → process-matched, monitored and qualified

CONSEQUENCE / LIMIT

- DEFINE → retain UNCCD dryland boundary
- MAP → locate process and pressure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIAGNOSE → susceptibility plus accelerator; RESPOND → process-matched, monitored and qualified.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

THE THAR LIES WEST OF THE ARAVALLI, BETWEEN THE

BOOK ESTIMATES OF DESERT AREA VARY WITH THE BOUNDARY

THE RANN OF KACHCHH IS TREATED AS AN EXTENSION

ECONOMY (MAJID)

BAJRA, PULSES, FODDER; LIVESTOCK IS CENTRAL; SCOPE FOR DATE-PALM

IT IS DESCRIBED AS THE WESTERN DRY REGION — ~29.7% OF INDIA (~97.85 MHA) IS DEGRADED (2018-19), UP

OPTIONAL SOURCE DEPTH

- The Thar lies west of the Aravalli, between the Indus and the Aravalli, spilling into Punjab, Haryana and northern Gujarat.
- Book estimates of desert area vary with the boundary used; retain them as source-era figures, not
- The Rann of Kachchh is treated as an extension of the desert; Jaisalmer and Barmer show the strongest aeolian (dune) topography.
- Economy (Majid): bajra, pulses, fodder; livestock is central; scope for date-palm, water-melon and rain-water harvesting.

ADVANCED DEBATE

- It is described as the Western Dry Region — an arid tract (Marwar/Mewar margins) with extreme temperature range.
- ~29.7% of India (~97.85 mha) is degraded (2018-19), up from 28.76% (2003-05) — a rising trend.
- Rajasthan has the largest desertified area; wind erosion drives 40%+ of its desertification.
- India targets Land Degradation Neutrality (LDN) by 2030 under the UNCCD; tools include Green India Mission and watershed/afforestation programmes.

LEGACY / NUANCE

- Thar = west of Aravalli, between Indus & Aravalli.
- Area figures vary by delimitation; western core is arid and margins semi-arid.
- Rann of Kachchh is linked to the wider arid region but is a distinct saline mudflat/marsh.
- Jaisalmer & Barmer = strongest aeolian (dune) topography.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels

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